# IPCC Tier 2 Cattle GHG Uncertainty Calculator

Technical Methodology: Equations, Monte Carlo Procedure, and Uncertainty Reporting










**Version 1.0**

Developed under the Global Methane Hub initiative

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# 1 Introduction

## 1.1 Purpose

The **IPCC Tier 2 Cattle GHG Uncertainty Calculator** is a web-based tool that enables national inventory compilers to quantify the uncertainty in cattle greenhouse gas (GHG) emission estimates produced using the IPCC Tier 2 methodology. The tool implements the Monte Carlo (Approach 2) uncertainty framework described in IPCC (2006) Volume 1, Chapter 3, applied to the full emission equation chain of IPCC (2006) Volume 4, Chapter 10 and Chapter 11 for pasture $N_2O$.

Under the Paris Agreement Enhanced Transparency Framework (ETF), national inventory reports are expected to include quantitative uncertainty estimates alongside emission totals. This tool automates that process, from parameter sampling to the formatted IPCC Table 3.3 output, for inventories of GHG emissions from one or more sub-categories of cattle.

## 1.2 Scope

The tool covers six emission sources for cattle (or buffalo):

1. **Enteric fermentation $CH_4$**: methane produced during enteric fermentation in the rumen
2. **Manure management $CH_4$**: methane from anaerobic decomposition of manure in storage systems
3. **Manure management $N_2O$ (direct)**: direct nitrous oxide from nitrogen in managed manure
4. **Manure management $N_2O$ (indirect)**: indirect $N_2O$ via atmospheric deposition and leaching from managed manure
5. **Pasture/range/paddock $N_2O$ (direct)**: direct $N_2O$ from dung and urine deposited on pasture
6. **Pasture/range/paddock $N_2O$ (indirect)**: indirect $N_2O$ from dung and urine on pasture

The tool supports multiple cattle sub-categories (e.g. dairy cows, beef bulls, heifers, growing males, calves) within a single inventory run, with results aggregated to national totals and to per-cattle-type subtotals (dairy, non-dairy), and with a per-sub-category breakdown that users can use to represent production systems, regions, or animal classes.

## 1.3 IPCC Tier Classification

The tool implements **Tier 2** methods throughout the emission chain. Tier 2 enteric fermentation outputs (GE, DE, CP) feed the manure $CH_4$ and $N_2O$ calculations, so the whole chain is classified as Tier 2. The uncertainty analysis itself follows the **Approach 2** (Monte Carlo) framework, which is the more accurate of the two IPCC uncertainty methods and is recommended when non-linear equation chains are involved.

# 2 Input Parameters

All parameters are entered via an Excel template (one row per parameter per cattle sub-category) and loaded into the tool. Each parameter requires a central value, an uncertainty specification (distribution type and bounds or percentage), and an optional IPCC reference.

## 2.1 What the defaults assume

Every IPCC default this tool ships is one cell of a much larger table. Reaching it means choosing a region, a productivity class, a climate and, for manure, a specific system variant. Those choices are listed below. **Where a choice does not describe the herd being reported, the parameters it governs must be replaced with country-specific values.**

| Choice | This tool uses | IPCC also publishes | Affects, and what to do otherwise |
|---|---|---|---|
| **Geography** | Africa | North America; Western Europe; Eastern Europe; Oceania; Latin America; Asia; Middle East; Indian subcontinent | `BW, MW, WG, Milk, Fat, MilkPR, pct_pregnant, DE, CP, hours`. The tool is built for developing-country inventory compilers. Country-specific values should replace these wherever they exist. |
| **Productivity class** | Low productivity | Regional aggregate (Milk 3.5, BW 260); high productivity (Milk 5.8, BW 250) | `BW, Milk, MilkPR, pct_pregnant, DE, CP, Bo, Ym`. Adopted 2026-09-11. Table 10.16A footnote 1 makes low productivity the Tier 1 default for other regions, and the low-productivity row's Pasture/Range feeding situation is the one that matches the activity coefficient the tool uses. Before this the defaults mixed the aggregate and non-dairy rows and described no animal IPCC published. |
| **Feeding situation** | Pasture / Range, flat terrain | Stall-fed (Ca 0); grazing large areas or hilly terrain (Ca 0.36) | `Ca`. Matches the low-productivity row, which Annex 10A.1 characterises as Pasture/Range. The tool has no feeding-situation input, so stall-fed or hilly-grazing herds must override the Ca parameter directly. |

| Choice | This tool uses | IPCC also publishes | Affects, and what to do otherwise |
|---|---|---|---|
| **Animal class for manure nitrogen** | Other Cattle, for every herd including dairy | Dairy Cow; Swine; Poultry; Other animals | `FRAC`. Table 10.22 publishes a separate Dairy Cow column and this tool does not yet use it, so a dairy herd receives the Other Cattle fractions. The two columns agree for lagoon, liquid slurry, daily spread, dry lot, deep bedding, aerobic treatment and digester. They differ for solid storage (0.45 against 0.30), covered solid storage (0.22 against 0.14) and composting (0.65 against 0.50). A DAIRY HERD USING SOLID STORAGE OR COMPOSTING SHOULD ENTER THE DAIRY COW VALUES FROM TABLE 10.22 rather than accept these. |
| **Climate, soils pathway** | Wet | Dry (EF3_PRP 0.002, EF4 0.005, Frac_LEACH_PRP 0); climate-aggregated (EF3_PRP 0.004, EF4 0.010) | `EF3_PRP, EF4, Frac_LEACH_PRP`. DRY-CLIMATE INVENTORIES MUST OVERRIDE THESE THREE. Keeping the wet-climate defaults in a dry climate overstates direct pasture N2O roughly threefold, overstates indirect N2O from deposition almost threefold, and reports a leaching pathway that IPCC treats as absent. |
| **Climate zone, manure methane** | Tropical, wherever the tool fills a methane conversion factor for you | Temperate and boreal reference values are on the Vocab sheet; the 2019 Refinement resolves ten zones where this tool resolves four, and its tropical-dry column mirrors tropical | `MCF`. Inventory_Metadata carries no climate field, so any MCF the tool fills in for you assumes a tropical climate. A TEMPERATE OR BOREAL INVENTORY MUST REPLACE EVERY GAP-FILLED MCF: tropical values run roughly two to five times the boreal ones for the same system. Values you enter yourself are used as given. |

| Choice | This tool uses | IPCC also publishes | Affects, and what to do otherwise |
|---|---|---|---|
| **Manure system variant** | One IPCC sub-type per manure system, named on every row of the manure-system table | Liquid slurry WITH versus without a natural crust; composting static pile versus in-vessel versus windrow; solid storage plain versus bulking agent versus additives; deep bedding with versus without mixing | `MCF`, `EF3`, `FRAC`. IPCC splits several systems into variants whose coefficients differ by a factor of two or more. Every coefficient on one of our rows comes from the single variant that row declares, so the row describes one real system rather than a blend. |
| **Species** | Cattle | Buffalo (Bo 0.10) | `Bo`, `Cp`, `ASH`. The tool is a cattle tool. Buffalo values differ and are not carried. |

`MCF`, `EF3` and `FRAC` above are the per-manure-system coefficient families rather than catalogue parameters.

### 2.1.1 Which IPCC variant each manure system models

IPCC splits several manure systems into variants whose coefficients differ substantially. Liquid slurry with a natural crust emits less methane and volatilises less nitrogen than the same slurry without one, and composting in a static pile is not the same system as composting in a vessel. This tool carries **one row per manure system**, and every coefficient on that row (the methane conversion factor, the direct $N_2O$ emission factor, and the volatilisation and leaching fractions) is read from the single IPCC variant named below, so the row describes a real system rather than a blend. **A herd running a different variant must override the coefficients on that row.**

| Manure system | IPCC variant modelled |
|---|---|
| `pasture` | PRP (Ch.11 pathway; MCF from 2006 Table 10.17 Pasture/Range/Paddock) |
| `daily_spread` | Daily spread |
| `solid_storage` | Solid storage (plain) |
| `solid_storage_covered` | Solid storage - Covered/compacted (2019R Table 10.17) |
| `dry_lot` | Dry lot |
| `deep_bedding` | Deep bedding, >1 month accumulation |
| `liquid_slurry` | Liquid/Slurry, with natural crust cover |
| `composting` | Composting - Static Pile (forced aeration) |

| Manure system | IPCC variant modelled |
|---|---|
| `lagoon` | Uncovered anaerobic lagoon |
| `anaerobic_digester` | Anaerobic digester, low leakage, high-quality industrial technology, open storage |
| `aerobic_treatment` | Aerobic treatment, forced aeration |
| `burned_for_fuel` | Burned for fuel |

Table 3 lists all parameters used in the calculation chain, their units, the IPCC equation in which they appear, the default value pre-loaded in the tool (sourced from the IPCC Guidelines), and an indication of potentially suitable probability distributions that can be applied in uncertainty analysis. Where the 2006 Guidelines and the 2019 Refinement give different defaults, both are shown.

**Table 3:** Input parameters for the IPCC Tier 2 cattle GHG calculation chain: including IPCC defaults and source citations.

| Variable | Definition | Unit | IPCC default | IPCC reference (equation / table) | Typical dist. |
|---|---|---|---|---|---|
| `N` | Number of animals (population) | head | *n/a* | Eq. 10.1 (Tier 1 totals); national activity data | Normal |
| `BW` | Mean live body weight | kg/head | 270 | Eq. 10.3; Annex 10A.1 (Africa, low productivity) | Normal |
| `Cfi` | Coefficient for $NE_m$ | MJ/kg$^{0.75}$/d | 0.386 (lact.) / 0.322 (non-lact.) / 0.370 (bulls) | Eq. 10.3; Table 10.4 (Updated) | Constant |
| `Tw` | Mean winter temperature | °C | 20 (inert) | Eq. 10.2 (cold-climate Cfi adjustment) | Constant |
| `Ca` | Activity coefficient | : | 0.00 (stall) / 0.17 (pasture flat) / 0.36 (large/hilly) | Eq. 10.4; Table 10.5 (Updated) | Constant |
| `WG` | Mean daily weight gain | kg/head/d | 0 (non-growing default) | Eq. 10.6 | Normal |
| `MW` | Mature body weight | kg/head | 300 | Eq. 10.6; Annex 10A.2 | Normal |

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*Table 3 continued*

| Variable | Definition | Unit | IPCC default | IPCC reference (equation / table) | Typical dist. |
|---|---|---|---|---|---|
| `C` | Sex/condition coefficient for $NE_g$ | : | 0.8 (female) / 1.0 (castrate) / 1.2 (bull) | Eq. 10.6 | Constant |
| `Milk` | Daily milk yield | kg/head/d | 1.2 (Africa, low productivity) | Eq. 10.8; Annex 10A.1 (Africa, low productivity) | Normal/Lognormal |
| `MilkFat` | Milk fat content | % | 4.3 (Africa, IPCC 2019R) | Eq. 10.8; Annex 10A.1 | Normal |
| `pct_pregnant` | Fraction of females pregnant / yr | : | 0.52 | Annex 10A.1 (Africa, low productivity) | Beta |
| `MilkPR` | Milk protein content | % | 3.6 ($\approx 1.9 + 0.4\times$ %Fat at Fat 4.3) | Eq. 10.33 (uses 6.38 milk-protein-to-N); Annex 10A.1–10A.3 | Normal |
| `WorkHours` | Daily working hours | h/head/d | 0 (no work) | Eq. 10.11 | Constant |
| `Cp` | Pregnancy coefficient | : | 0.10 (cattle) | Eq. 10.13; Table 10.7 (Updated) | Constant |

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*Table 3 continued*

| Variable | Definition | Unit | IPCC default | IPCC reference (equation / table) | Typical dist. |
|---|---|---|---|---|---|
| DE_pct | Digestible energy (% of GE) | % | 51.0 (Africa, low productivity) | Eq. 10.14–16; Annex 10A.1 (Africa, low productivity) | Normal |
| Ym | Methane conversion factor | % | 6.5 (lactating dairy cows, the catalogue default); resolved per subcategory: 7.0 non-dairy and multi-purpose, 4.0 feedlot under 2019R; 6.5 and 3.0 respectively under 2006 | Eq. 10.21; Table 10.12 (Updated), footnote 4 | Normal/PERT |
| UE | Urinary energy (fraction of GE) | : | 0.04 (cattle) | Eq. 10.24 (footnote) | Constant |

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*Table 3 continued*

| Variable | Definition | Unit | IPCC default | IPCC reference (equation / table) | Typical dist. |
|---|---|---|---|---|---|
| `ASH` | Ash content of manure (fraction of DMI) | : | 0.08 (cattle; IPCC example 0.06 is pig-specific) | Eq. 10.24 | Constant |
| `MMS_fractions` | Fraction of manure in each MMS (deterministic; constrained to sum to 1 per sub-category). Optional per-row `lower_fraction` / `upper_fraction` / `distribution_fraction` columns on the Manure_Management sheet enable per-iteration sampling within the user's bounds, renormalised to preserve the simplex; sampled fractions surface as `fraction_<mms>` in the sensitivity tornado. | : | country-specific (Worksheet 10.A-9) | Eq. 10.23 | Deterministic (default) or per-MMS PERT/Normal/etc. with renormalisation |

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*Table 3 continued*

| Variable | Definition | Unit | IPCC default | IPCC reference (equation / table) | Typical dist. |
|---|---|---|---|---|---|
| `Bo` | Maximum $CH_4$ producing capacity | $m^3$ $CH_4$/kg VS | 0.13 (2019R "Other regions, low productivity" cattle); 2006 Africa = 0.10 | Eq. 10.23; Table 10.16(a) (Updated) | Normal |
| `MCF` | Methane conversion factor (MMS) | : | per-MMS & climate (e.g. 80% lagoon tropical) | Eq. 10.23; Table 10.17 (Updated) | Normal/PERT |
| `EF3` | Direct-$N_2O$ EF (per MMS) | kg $N_2O$-N/kg N | per-MMS (e.g. 0.010 solid storage) | Eq. 10.25; Table 10.21 (Updated) | Lognormal/PERT |
| `Frac_GasMS` | Fraction N volatilised (MMS) | : | per-MMS, e.g. 0.45 solid storage (2019R) | Eq. 10.26; Table 10.22 (Updated) | PERT (preferred, IPCC ranges are asymmetric) / Normal |

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*Table 3 continued*

| Variable | Definition | Unit | IPCC default | IPCC reference (equation / table) | Typical dist. |
|---|---|---|---|---|---|
| `EF4` | EF for atmospheric deposition | kg $N_2O$-N/(kg $NH_3$-N+$NO_x$-N) | **0.014** (2019R wet climate, the shipped default); aggregated 0.010, dry 0.005; 2006 = 0.010 | Eq. 10.27; Vol.4 Ch.11 Table 11.3 (Updated) | PERT |
| `Frac_LeachMS` | Fraction N leached (MMS) | : | per-MMS, e.g. 0.02 solid storage | Eq. 10.28; Table 10.22 (Updated) | PERT |
| `EF5` | EF for leaching/runoff | kg $N_2O$-N/kg N | 0.011 (2019R); 2006 = 0.0075 | Eq. 10.29; Vol.4 Ch.11 Table 11.3 (Updated) | PERT |
| `Frac_PRP` | Fraction of dung and urine deposited on pasture | : | country-specific | Vol.4 Ch.11 Eq. 11.5 ($MS_{(T,PRP)}$) | Beta / PERT (assignable via the `distribution` column) |

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*Table 3 continued*

| Variable | Definition | Unit | IPCC default | IPCC reference (equation / table) | Typical dist. |
|---|---|---|---|---|---|
| `EF3_PRP` | Direct $N_2O$ EF, pasture (cattle/poultry/pigs) | kg $N_2O$-N/kg N | **0.006** (2019R wet climate, the shipped default); aggregated 0.004, dry 0.002; 2006 = 0.02 | Vol.4 Ch.11 Eq. 11.1; Table 11.1 (Updated) | PERT |
| `Frac_GasPRP` | Fraction N volatilised (PRP) | : | 0.21 (2019R FracGASM); 2006 = 0.20 | Vol.4 Ch.11 Eq. 11.9; Table 11.3 (Updated) | PERT (preferred, IPCC ranges are asymmetric) / Normal |
| `Frac_LeachPRP` | Fraction N leached (PRP) | : | 0.24 (2019R wet); 2006 = 0.30; dry = 0 | Vol.4 Ch.11 Eq. 11.10; Table 11.3 (Updated) | PERT |

# 3 IPCC Tier 2 Emission Equations

This section presents the complete equation chain implemented in the tool, following IPCC (2006) Volume 4, Chapter 10. All equations are evaluated for each cattle sub-category independently; results are then aggregated to national totals by summing across sub-categories.

## 3.1 Net Energy for Maintenance

The daily net energy requirement for maintenance ($NE_m$, MJ head$^{-1}$ day$^{-1}$) is given by IPCC Eq. 10.3:

$$NE_m = C_{fi,\text{adj}} \times BW^{0.75}$$

where $BW$ is the mean live body weight (kg head$^{-1}$) and $C_{fi}$ is a sex/condition coefficient (MJ kg$^{-0.75}$ day$^{-1}$). For open-lot cattle in colder climates the 2006 Guidelines (Vol.4 Ch.10, retained unchanged in the 2019 Refinement) prescribe a linear adjustment of $C_{fi}$: IPCC Eq.~10.2, "Coefficient for calculating net energy for maintenance'', based on the mean daily winter temperature:

$$C_{fi,\text{adj}} = C_{fi} + 0.0048 \times (20 - T_w) \quad \text{when } T_w < 20°\text{C}$$

otherwise $C_{fi,\text{adj}} = C_{fi}$. This adjustment increases the maintenance energy requirement by approximately 0.48% per degree below 20°C. The same linear formula is implemented in the IPCC Inventory Software (v2.95).

## 3.2 Net Energy for Activity

The net energy for physical activity ($NE_a$, MJ head$^{-1}$ day$^{-1}$) is given by IPCC Eq. 10.4:

$$NE_a = C_a \times NE_m$$

where $C_a$ is a dimensionless activity coefficient that accounts for the terrain and movement patterns of the animal (e.g. 0.00 for housed animals, 0.17 for animals on flat pasture, 0.36 for large areas or hilly terrain).

## 3.3 Net Energy for Growth

The net energy required for growth ($NE_g$, MJ head$^{-1}$ day$^{-1}$) is given by IPCC Eq. 10.6:

$$NE_g = 22.02 \times \left(\frac{BW}{C \times MW}\right)^{0.75} \times WG^{1.097}$$

where $WG$ is the mean daily weight gain (kg head$^{-1}$ day$^{-1}$), $MW$ is the mature body weight (kg),

and $C$ is a sex/physiological condition coefficient (0.8 for females, 1.0 for castrates, 1.2 for bulls). $NE_g = 0$ when $WG \leq 0$.

### 3.4 Net Energy for Lactation

The daily net energy for milk production ($NE_l$, MJ head$^{-1}$ day$^{-1}$) is given by IPCC Eq. 10.8:

$$NE_l = \text{Milk} \times (1.47 + 0.40 \times \text{MilkFat})$$

where Milk is the daily milk yield (kg head$^{-1}$ day$^{-1}$) and MilkFat is the fat content (%). For non-lactating sub-categories, set Milk = 0.

### 3.5 Net Energy for Work

The net energy for work ($NE_w$, MJ head$^{-1}$ day$^{-1}$) is given by IPCC Eq. 10.11 (used when cattle perform physical work, e.g. traction or load-carrying):

$$NE_w = 0.10 \times NE_m \times \text{WorkHours}$$

where WorkHours is the mean number of hours of work per day. Set to 0 when animals do not perform work.

### 3.6 Net Energy for Pregnancy

The net energy allocated to fetal growth during pregnancy ($NE_p$, MJ head$^{-1}$ day$^{-1}$) is given by IPCC Eq. 10.13:

$$NE_p = C_p \times NE_m$$

where $C_p$ is a coefficient for the energy cost of pregnancy (0.10 for cattle). The parameter `pct_pregnant` can pro-rate $C_p$ when not all females in a sub-category are pregnant in the same year ($0 \leq$ pct_pregnant $\leq 1$). Includes pregnant heifers that have not yet calved. The earlier name `pct_calving` is accepted as an alias on upload.

### 3.7 Energy Ratios: REM and REG

The ratio of net energy for maintenance to digestible energy consumed ($REM$) is given by IPCC Eq. 10.14:

$$REM = 1.123 - (4.092 \times 10^{-3} \times DE) + (1.126 \times 10^{-5} \times DE^2) - \frac{25.4}{DE}$$

The ratio of net energy for growth to digestible energy consumed ($REG$) is given by IPCC Eq. 10.15:

$$REG = 1.164 - (5.160 \times 10^{-3} \times DE) + (1.308 \times 10^{-5} \times DE^2) - \frac{37.4}{DE}$$

where $DE$ is the digestible energy expressed as a percentage of gross energy. Both equations are empirical fits derived from ruminant data and are only valid over the digestibility range observed for ruminant diets. The 2019 Refinement (Vol.4 Ch.10, §10.2) documents that range as **45–85%** (45–55% for crop by-products and rangelands; 55–80% for managed pasture and forage-based diets; 72–85% for grain-based feedlot diets). Outside this range the fits extrapolate poorly: $REG$ passes through zero near $DE \approx 37.6\%$, and below that the $-37.4/DE$ term dominates and drives $REG$ negative. Because the gross-energy equation (below) divides by $REM$ and $REG$, an unconstrained low-$DE$ draw makes gross energy, and hence emissions, diverge or even turn negative.

To keep every Monte Carlo iteration inside the equations' valid domain, sampled $DE$ values are **clamped to** $[45, 85]\%$ at the sampling stage (`generate_mc_samples`). This prevents non-physical iterations (the extreme positive and negative emission totals that would otherwise stretch the output histogram and pull the simulated mean above the $97.5^{\text{th}}$ percentile) without materially shifting the central estimate, since the clamp only trims the small fraction of the $DE$ distribution that falls outside the documented ruminant range. The earlier $DE > 0$ check is retained as defence-in-depth.

### 3.8 Gross Energy Intake

The mean daily gross energy intake ($GE$, MJ head$^{-1}$ day$^{-1}$) is given by IPCC Eq. 10.16:

$$GE = \frac{(NE_m + NE_a + NE_l + NE_w + NE_p)/REM + NE_g/REG}{DE/100}$$

### 3.9 Two Routes to Gross Energy: Modelled and Measured

Equation 10.16 above infers intake from what the animal does. Its inputs are animal performance: body weight, growth, milk yield, pregnancy and activity. This is the route the tool takes by default, and the only route earlier versions offered.

Many datasets work in the opposite direction. They measure feed intake and feed quality, and so already hold $GE$, or hold dry matter intake ($DMI$) and can convert. Such datasets frequently carry no milk yield or weight gain at all, because nothing in their method required inferring intake. Measured intake is not a lesser input than modelled intake; where it exists it is generally the stronger evidence.

The tool therefore accepts $GE$ or $DMI$ as optional inputs, **per sub-category**:

1. If a usable $GE$ is supplied for a sub-category, it is used directly and Equations 10.3 to 10.16 are not evaluated for that sub-category.
2. Otherwise, if a usable $DMI$ is supplied, $GE = DMI \times 18.45$.
3. Otherwise $GE$ is derived from Equation 10.16 exactly as above.

A value is "usable" only if it is present, finite and strictly positive. Blank, missing, zero and negative all mean "not supplied", so a zero intake is never mistaken for a measurement. Because the choice is made per sub-category, one inventory may mix measured and modelled groups, and an inventory that supplies neither is numerically unchanged from earlier versions of the tool.

**$GE$ takes precedence over $DMI$.** A supplied $GE$ carries the dataset's own measured dietary energy density, whereas converting $DMI$ imposes the IPCC figure of 18.45 MJ kg$^{-1}$ dry matter. Where both are given, using $GE$ preserves the user's measurement.

### 3.9.1 What a measured intake does not replace

Supplying $GE$ bypasses the energy balance only. Three groups of inputs remain necessary, and this is a frequent source of confusion:

- **Digestible energy** $DE$ is still required. With $GE$ supplied it no longer affects enteric methane, but Equation 10.24 still uses it to set the indigestible share of the feed that becomes volatile solids, and hence manure methane.
- **Milk yield, milk protein content and weight gain** are still required. Equations 10.32 and 10.33 obtain nitrogen excretion by subtracting the nitrogen retained in milk and in liveweight gain from the nitrogen eaten. These inputs therefore continue to determine direct and indirect $N_2O$ from both managed manure and pasture.
- **Crude protein** $CP$ is still required, for the same nitrogen-intake term.

### 3.9.2 A residual the method itself carries

Equations 10.24 and 10.32 divide by the IPCC constant of 18.45 MJ kg$^{-1}$ dry matter regardless of the user's own feed. A dataset whose measured energy density differs, for instance the 16.5 to 17.6 MJ kg$^{-1}$ typical of tropical forage-based systems, will therefore see that constant applied in the volatile-solids and nitrogen-intake terms even while its own $GE$ is used verbatim for enteric methane. This is a property of the IPCC equations, not a defect in the implementation, and it is documented here so it is not mistaken for one.

### 3.9.3 Reporting which route was used

Every run records, for each sub-category, whether gross energy was modelled or measured. The label appears in the results table in the application, in the run-settings sheet of the Excel export, and in the Word report. Where an inventory mixes routes, the split is stated explicitly rather than a single method being implied.

## 3.10 Enteric Fermentation Emission Factor

The per-head per-day methane emission factor for enteric fermentation ($EF_1$, kg $CH_4$ head$^{-1}$ day$^{-1}$) is given by IPCC Eq. 10.21:

$$EF_1 = \frac{GE \times (Y_m/100)}{55.65}$$

where $Y_m$ is the methane conversion factor expressed as a percentage of gross energy converted to methane (the /100 converts $Y_m$ to a fraction) and 55.65 MJ kg$^{-1}$ is the energy content of methane. The annual per-sub-category enteric emission is:

$$E_{\text{enteric}} = EF_1 \times N \times 365/1000 \quad (\text{tonnes CH}_4\ \text{yr}^{-1})$$

### 3.11 Nitrogen Excretion

The daily nitrogen excretion rate ($N_{\text{ex}}$, kg N head$^{-1}$ day$^{-1}$) follows IPCC Eq. 10.31:

$$N_{\text{ex}} = N_{\text{intake}} - N_{\text{retained}}$$

Nitrogen intake from feed is:

$$N_{\text{intake}} = \frac{GE}{18.45} \times \frac{\text{CP\%}}{6.25 \times 100}$$

where CP% is the crude-protein content of the diet (the template column is labelled `CP`; the value supplied is the same percentage value used here).

Nitrogen retained in body tissue and milk are calculated separately. For milk, the nitrogen content uses the **Jones factor for casein (6.38)**: distinct from the generic protein factor (6.25) used for feed and body tissue. The 6.38 factor and the 6.25 factor both appear inside IPCC 2006 Vol.4 Ch.10 Eq. 10.33 (N retention rates for cattle); the same equation is retained, with updated form, in the 2019 Refinement (also numbered 10.33):

$$N_{\text{milk}} = \frac{\text{Milk} \times \text{MilkProtein\%}}{6.38 \times 100}$$

For growing animals, IPCC Eq. 10.33 adds a body-tissue retention term:

$$N_{\text{growth}} = \frac{WG \times (268 - 7.03 \times NE_g/WG)}{1000 \times 6.25}$$

so that $N_{\text{retained}} = N_{\text{milk}} + N_{\text{growth}}$. For non-growing sub-categories ($WG \leq 0$) the growth term is set to zero.

The annual nitrogen excreted per sub-category ($N_{\text{ex,annual}}$, kg N yr$^{-1}$) is:

$$N_{\text{ex,annual}} = N_{\text{ex}} \times N \times 365$$

### 3.12 Volatile Solids and Manure $CH_4$

Total manure $CH_4$ emissions are given by IPCC 2006 Vol.4 Ch.10 Eq. 10.22 (the top-level emissions equation, EF × population × $10^{-6}$). The per-head per-year emission factor is given by Eq. 10.23:

$$EF_{\mathrm{mm,CH_4}} = VS \times 365 \times B_o \times 0.67 \times \sum_s (MS_s \times MCF_s) \quad \text{(Eq. 10.23)}$$

The daily volatile-solid excretion rate $VS$ (kg dry-matter head$^{-1}$ day$^{-1}$) that feeds Eq. 10.23 is given by IPCC Eq. 10.24:

$$VS = [GE \times (1 - DE/100) + (UE \times GE)] \times \left(\frac{1 - \mathrm{ASH}}{18.45}\right) \quad \text{(Eq. 10.24)}$$

where $UE$ is the fraction of GE excreted as urinary energy (default 0.04 for cattle, IPCC Eq. 10.24 footnote) and ASH is the ash content of manure (default 0.08 for cattle).

where $B_o$ is the maximum methane producing capacity (m$^3$ $CH_4$ kg$^{-1}$ VS), 0.67 is the conversion factor from m$^3$ $CH_4$ to kg $CH_4$ (the density of methane at standard conditions, kg m$^{-3}$), $MS_s$ is the fraction of manure managed in system $s$, and $MCF_s$ is the methane conversion factor for system $s$ (fraction of $B_o$). The 2019 Refinement Table 10.17 (Updated) provides MCF defaults across 11 manure management systems: eight inherited from the 2006 Guidelines plus four added in 2019 (anaerobic digester / biogas, aerobic treatment, solid storage – covered/compacted, and burned for fuel). Default $B_o$ values for cattle are given in Table 10.16a. Per-MMS uncertainty in $MCF$ and in the manure fractions $MS_s$ is sampled independently, when the user supplies optional `lower_fraction` / `upper_fraction` / `distribution_fraction` columns on the Manure_Management sheet, each Monte Carlo iteration's $MS_s$ row is renormalised after sampling so the simplex constraint $\sum_s MS_s = 1$ is preserved exactly.

### 3.13 Manure Management $N_2O$ (Direct)

Direct $N_2O$ emissions from manure management are given by IPCC 2006 Vol.4 Ch.10 Eq. 10.25 (the indirect-pathway losses are calculated separately in Eq. 10.26–10.29; see next sub-section):

$$E_{\mathrm{direct,mm}} = N_{\mathrm{ex,annual}} \times EF_3 \times \frac{44}{28} \quad (\text{kg } N_2O \text{ yr}^{-1})$$

where $EF_3$ is the emission factor for direct $N_2O$ (kg $N_2O$-N kg$^{-1}$ N) and 44/28 converts $N_2O$-N to $N_2O$. For multiple MMS systems, $EF_3$ is computed as the weighted average of per-MMS factors:

$$EF_3 = \sum_s MS_s \times EF_{3,s}$$

### 3.14 Manure Management $N_2O$ (Indirect)

Indirect $N_2O$ from managed manure arises from two pathways. Each pathway is split between an N-loss equation (in Vol.4 Ch.10) and an $N_2O$ equation (in Vol.4 Ch.10, drawing EFs from Vol.4 Ch.11 Table 11.3):

**Atmospheric deposition** of volatilised N: Frac_GasMS from Eq. 10.26 multiplied by $EF_4$ in Eq. 10.27:

$$E_{\text{vol,mm}} = N_{\text{ex,annual}} \times \text{Frac_GasMS} \times EF_4 \times \frac{44}{28}$$

**Leaching and runoff**: Frac_LeachMS from Eq. 10.28 multiplied by $EF_5$ in Eq. 10.29:

$$E_{\text{leach,mm}} = N_{\text{ex,annual}} \times \text{Frac_LeachMS} \times EF_5 \times \frac{44}{28}$$

where $EF_4$ (kg $N_2O$-N kg$^{-1}$ $NH_3$-N + $NO_x$-N volatilised) and $EF_5$ (kg $N_2O$-N kg$^{-1}$ N leached/runoff) are from IPCC Vol.4 Ch.11 Table 11.3. In the 2006 Guidelines $EF_4 = 0.010$ (range 0.002–0.05) and $EF_5 = 0.0075$ (range 0.0005–0.025). The 2019 Refinement keeps the aggregated $EF_4 = 0.010$ (range 0.002–0.018) and additionally provides climate-disaggregated values of 0.014 in wet climates (range 0.011–0.017) and 0.005 in dry climates (range 0.000–0.011); $EF_5$ is updated to 0.011 (range 0.000–0.020) with no climate disaggregation. Frac_GasMS and Frac_LeachMS are MMS- and country- or climate-specific fractions (defaults in Vol.4 Ch.10 Tables 10.22 and 10.23 of the 2019 Refinement).

### 3.15 Pasture/Range/Paddock $N_2O$ (Direct)

The nitrogen deposited on pasture by grazing animals ($F_{\text{PRP}}$) is the per-sub-category form of IPCC Vol.4 Ch.11 Eq.~11.5 (N in urine and dung deposited by grazing animals on PRP):

$$F_{\text{PRP}} = N_{\text{ex,annual}} \times \text{Frac_PRP}$$

where Frac_PRP is the fraction of total annual N excretion deposited on pasture (denoted $MS_{(T,\text{PRP})}$ in IPCC notation). Direct $N_2O$ from pasture/range/paddock falls under managed soils (IPCC Volume 4, Chapter 11), not Chapter 10. It is given by IPCC Eq.~11.1 (with $F_{\text{PRP}}$ as the input N source):

$$E_{\text{direct,PRP}} = F_{\text{PRP}} \times EF_{3,\text{PRP}} \times \frac{44}{28}$$

where $EF_{3,\text{PRP}}$ is the emission factor for direct $N_2O$ from PRP given in IPCC Vol.4 Chapter 11, Table 11.1. The 2006 value is 0.02 kg $N_2O$-N kg$^{-1}$ N (cattle, poultry, pigs combined). The 2019 Refinement updates this to $EF_{3PRP,CPP}$ for cattle, poultry and pigs with an aggregated default of 0.004 kg $N_2O$-N kg$^{-1}$ N (range 0.000–0.014). When the climate is known, the disaggregated values are 0.006 in wet climates (range 0.000–0.027) and 0.002 in dry climates (range 0.000–0.007).

### 3.16 Pasture $N_2O$ (Indirect)

Indirect $N_2O$ from pasture deposition follows IPCC Vol.4 Ch.11 Eq. 11.9 (volatilisation pathway) and Eq. 11.10 (leaching/runoff pathway), using the same pathway structure as managed manure:

$$E_{\text{indirect,PRP}} = F_{\text{PRP}} \times (\text{Frac_GasPRP} \times EF_4 + \text{Frac_LeachPRP} \times EF_5) \times \frac{44}{28}$$

On the soil side of Eq.~11.9/11.10 the same role played by Frac_GasMS and Frac_LeachMS for managed manure is played by FracGASM and FracLEACH-(H) in IPCC Vol.4 Ch.11 Table 11.3: Frac_GasPRP corresponds to FracGASM (the fraction of applied/deposited N volatilised as $NH_3$ and $NO_x$) and Frac_LeachPRP corresponds to FracLEACH-(H) (the fraction leached/runoff in regions with annual precipitation exceeding evapotranspiration).

Per IPCC Vol.4 Ch.11, where evapotranspiration exceeds precipitation, leaching may be assumed not to occur: in the tool, `Frac_LeachPRP` may be set equal to zero in those climate cases.

### 3.17 GWP-Based $CO_2$eq Conversion

All emission pathways are converted to $CO_2$ equivalents using Global Warming Potentials (GWP). The tool supports three IPCC assessment report baselines:

| **Basis** | **$CH_4$ GWP** | **$N_2O$ GWP** | **Reference** |
|---|---|---|---|
| AR4 (100-yr) | 25 | 298 | IPCC AR4 WG1, Ch.2 Table 2.14 |
| AR5 (100-yr) | 28 | 265 | IPCC AR5 WG1, Ch.8 Table 8.7 |
| AR6 (100-yr, non-fossil $CH_4$) | 27.0 | 273 | IPCC AR6 WG1, Ch.7 Table 7.15 (and Table 7.SM |

Total $CO_2$eq = (total $CH_4$ tonnes $\times$ $GWP_{CH_4}$) + (total $N_2O$ tonnes $\times$ $GWP_{N_2O}$).

# 4 Monte Carlo Uncertainty Analysis

## 4.1 IPCC Approach 2 Rationale

The IPCC (2006) Volume 1, Chapter 3 defines two approaches to quantifying inventory uncertainty:

- **Approach 1 (error propagation):** Combines individual parameter uncertainties using simple algebraic rules. Suitable when the emission equation is a product or sum of independent terms. Underestimates uncertainty when equations are non-linear or inputs are correlated.
- **Approach 2 (Monte Carlo simulation):** Propagates uncertainty through the full equation chain by repeatedly sampling all parameters from their distributions. Exact for any equation structure. Recommended for Tier 2 livestock inventories because the GE equation chain is non-linear in $BW$, $WG$, and $DE_{\%}$.

This tool implements Approach 2. The simulation samples all input parameters simultaneously for each iteration, evaluates the full equation chain from $NE_m$ through to total $CO_2$eq, and records the output. The resulting distribution of outputs characterises the uncertainty.

## 4.2 Probability Distributions

Each parameter is assigned a probability distribution reflecting the state of knowledge about its true value. The tool supports:

| Distribution | Parameterisation | Typical use |
|---|---|---|
| Normal | mean $\mu$, SD $\sigma$ | Body weight, milk yield, population counts with symmetric survey error |
| Positive-Normal | Normal truncated at zero | Non-negative symmetric parameters where negative draws are implausible |
| Lognormal | log-mean, log-SD (fitted from mean and CV%) | Strictly positive, right-skewed; milk yield in heterogeneous herds; some EFs |
| Beta | rescaled Beta($\alpha, \beta$) matching the supplied mean and bounds | Fractions in $[0, 1]$ with mode strictly in the interior; pct_pregnant, Cp |
| Truncated-Normal [0,1] | Normal clipped to $[0, 1]$ | Fractions in $[0, 1]$ when a symmetric error around the mean is appropriate |
| PERT | minimum, most-likely, maximum | Emission factors with expert-elicited bounds; EF3_PRP, EF4, EF5, Bo, MCF |
| Triangular | minimum, most-likely, maximum | Similar to PERT but with sharper tails; Ca, C_growth |
| Uniform | lower, upper | Parameters with only the plausible range known; MMS fractions |
| Constant | fixed value | Parameters treated as known exactly (uncertainty_pct = 0) |

When the user supplies `uncertainty_pct` (a symmetric 95% half-width as a percentage of the mean), the tool derives the distribution parameters as:

$$\sigma = \frac{\text{uncertainty_pct} \times \mu}{100 \times 1.96}$$

for a normal distribution, or equivalent transformations for lognormal and PERT.

### 4.3 Correlated Sampling

When parameters are correlated (e.g. body weight and mature weight are positively correlated; diet quality and methane conversion factor may be negatively correlated), ignoring correlation underestimates or overestimates combined uncertainty. The tool applies the same correlated-sampling procedure to every path that supplies a correlation matrix: the structural-defaults preset, a manually uploaded matrix, and the matrix estimated automatically from a time-series upload all flow through the same algorithm.

**Procedure** (per IPCC Volume 1 Chapter 3 §3.2.3.2):

1. For each parameter $j$, draw $n_{\text{iter}}$ independent samples from the parameter's own marginal distribution (normal, PERT, lognormal, beta, triangular, uniform, or constant, exactly as specified in the template).
2. Reorder the columns of the resulting $n_{\text{iter}} \times p$ sample matrix so that the realised Spearman rank correlation between columns matches the target matrix $\mathbf{\Sigma}$ (implemented via `mc2d::cornode`, which performs the restricted-pairing operation IPCC §3.2.3.2 cites under "restricted pairing methods that are included in many software packages'').

This procedure is distribution-free: the marginal shapes are preserved exactly, regardless of whether the user has mixed normal / lognormal / PERT / beta parameters in the same correlation block. The target $\mathbf{\Sigma}$ is interpreted as a Spearman rank correlation matrix throughout, the same convention the time-series matrix is computed under (Spearman rank correlation of the user's annual values, after detrending; see Section "Correlation Structure'').

If $\mathbf{\Sigma}$ is not positive definite (e.g. a manually entered matrix that turns out to be inconsistent), the tool projects it onto the nearest positive-definite correlation matrix via `Matrix::nearPD(corr = TRUE)` and emits a warning whenever any individual entry is shifted by more than 0.01. The tool does *not* silently fall back to uncorrelated sampling.

The tool supports both within-block and across-block correlations. The default scope is independence; a single unified correlation matrix spanning AD + coefficients is used whenever cross-block pairs are specified.

### 4.4 Simulation Procedure

For each Monte Carlo iteration $i = 1, \ldots, n_{\text{iter}}$:

1. Sample all parameters from their distributions (with correlation structure if supplied).
2. For each cattle sub-category $k$:
   a. Compute $NE_m^{(k)}$, $NE_a^{(k)}$, $NE_g^{(k)}$, $NE_l^{(k)}$, $NE_w^{(k)}$, $NE_p^{(k)}$.
   b. Compute $REM^{(k)}$ and $REG^{(k)}$ from $DE_{\%}^{(k)}$.
   c. Compute $GE^{(k)}$, $EF_1^{(k)}$, $N_{\text{ex}}^{(k)}$, $VS^{(k)}$.
   d. Compute emission totals for all six pathways.
3. Sum across all sub-categories to obtain national pathway totals and total $CO_2$eq.
4. Record all outputs for iteration $i$.

After all iterations, compute summary statistics from the distribution of recorded outputs.

### 4.5 Recommended Iteration Count and Convergence

Monte Carlo estimates converge as $n_{\text{iter}} \to \infty$. For practical use:

- **1,000 iterations**: suitable only for a quick test run to verify the model runs without errors. Do not report results from 1,000 iterations.
- **10,000 iterations**: minimum recommended for any result intended for use. Convergence is

generally adequate for inventories with up to 5 sub-categories and 30 parameters.

- **25,000–30,000 iterations**: recommended for final reporting, especially when correlations are enabled or the inventory has many sub-categories or asymmetric distributions.
- **50,000 iterations**: high-precision run; use when the 95% CI bounds need to be stable to two significant figures.

**Convergence check:** Re-run with a different random seed (e.g. 123 and 456). If the 95% MoE changes by more than 1–2 percentage points, increase the number of iterations.

# 5 Uncertainty Metrics

The following metrics are reported for each emission pathway and for the total $CO_2$eq:

## 5.1 95% Confidence Interval

The 95% CI is defined as the 2.5th and 97.5th percentiles of the Monte Carlo output distribution:

$$\mathrm{CI}_{95} = [Q_{0.025},\ Q_{0.975}]$$

This is a simulation-based (quantile) interval: it does not assume a normal distribution and is therefore valid for asymmetric emission distributions.

## 5.2 Margin of Error (the IPCC reporting metric)

The margin of error (MoE%) is the half-width of the 95% CI expressed as a percentage of the mean:

$$\mathrm{MoE}\% = \frac{Q_{0.975} - Q_{0.025}}{2 \times \bar{x}} \times 100$$

**MoE% is the metric IPCC Vol.1 Ch.3 Table 3.3 requires** ("% uncertainty" in that table is defined as the half-width of the 95% CI divided by the mean). It is the headline number reported in the user-facing value boxes, in the IPCC Report tab, and in the Excel / Word downloads. Because MoE is derived from the empirical 2.5% / 97.5% quantiles of the Monte Carlo output, it correctly captures distribution skewness: for a perfectly normal distribution it reduces to $\approx 1.96 \times \mathrm{CV}\%$, but for skewed distributions (lognormal EFs, PERT-bounded fractions, manure-$N_2O$ sums) MoE and $1.96 \cdot \mathrm{CV}$ differ.

## 5.3 Coefficient of Variation (kept for reference, not reported as headline)

The coefficient of variation (CV%) expresses uncertainty relative to the mean:

$$\mathrm{CV}\% = \frac{\sigma}{\bar{x}} \times 100$$

where $\sigma$ is the standard deviation and $\bar{x}$ is the mean. CV% is **still computed internally** for every emission source (kept in the underlying `calc_all_uncertainty()` data frame for advanced reproducibility) but it is no longer surfaced in any user-facing display: IPCC Annex 7 expects MoE%, not CV%. It is undefined (reported as NA) when the mean emission is zero (e.g. pasture $N_2O$ for fully housed systems).

## 5.4 IPCC Table 3.3 Format

IPCC (2006) Volume 1, Table 3.3 requires reporting uncertainty for each emission category as follows:

| Source / Sink Category | Emission (t $CH_4$ or t $N_2O$) | AD Uncertainty (95% MoE %) | EF Uncertainty (95% MoE %) | Combined Uncertainty (95% MoE %) |
|---|---|---|---|---|
| Enteric fermentation $CH_4$ | ... | MoE% from AD-only run | MoE% from EF-only run | MoE% from combined run |
| ... | | | | |

The tool generates this table automatically after each simulation run. The Excel download labels each uncertainty column explicitly as "(95% MoE %)" so there is no ambiguity with CV.

# 6 Uncertainty Decomposition

To separate the contributions of activity data (AD) uncertainty and emission factor (EF) uncertainty to the combined uncertainty, the tool performs three simulation runs:

1. **Combined run**: all parameters varied according to their distributions. This is the primary result.
2. **AD-only run**: emission factors fixed at their central values ($\sigma = 0$); only AD parameters vary. 95% MoE from this run = AD uncertainty contribution.
3. **EF-only run**: AD parameters fixed at their central values; only emission factors vary. 95% MoE from this run = EF uncertainty contribution.

This approach correctly handles multi-group inventories by running all three simulations through the full multi-sub-category equation chain and aggregating to national totals before computing metrics.

The decomposition reveals whether uncertainty reduction efforts should focus on improving animal population surveys (AD) or on refining emission factor estimates (EF).

*Implementation note.* The AD-only run pins ALL coefficient-side variance, not only the Parameters-sheet rows. The per-MMS sample matrices used for MCF, EF3, Frac_GasMS, Frac_LeachMS, and (when supplied) the MMS-allocation $MS_s$ fractions are nulled out for the AD-only sub-run so iteration-to-iteration variation in those quantities cannot leak into the AD-only result. As a result, AD-only CV equals CV(N) for every emission source in a single-sub-category inventory and varies across sources only through the Herfindahl-like concentration of each source over multiple sub-categories.

# 7 Sensitivity Analysis

Sensitivity analysis identifies which input parameters have the greatest influence on the total emission estimate. The tool implements two complementary methods.

## 7.1 Standardised Regression Coefficients (SRC)

SRC is based on fitting a multiple linear regression of the standardised output on all standardised inputs:

$$Y^* = \beta_0 + \sum_j \beta_j X_j^* + \varepsilon$$

where $Y^* = (Y - \bar{Y})/\sigma_Y$ and $X_j^* = (X_j - \bar{X}_j)/\sigma_{X_j}$. The SRC for parameter $j$ is the coefficient $\beta_j$ from this regression.

Properties: - A positive SRC indicates that higher values of the parameter produce higher emissions. - The magnitude $|\beta_j|$ reflects the relative importance of parameter $j$. - SRC values range from approximately $-1$ to $+1$ when a single parameter dominates. - The $R^2$ of the regression indicates what fraction of the total variance in the output is explained by the linear model. High $R^2$ ($> 0.9$) indicates SRC is a reliable measure; low $R^2$ suggests non-linearity. - SRC requires only a single regression fit, making it computationally efficient for any number of parameters.

## 7.2 Tornado Chart Interpretation

Each parameter label on the tornado chart is formatted as `<parameter> (<sub_category>)`: for example `Ym (DINT_cow)`, `BW (DINT_heif)`, `MCF_solid_storage (DINT_cow)`. The sub-category in parentheses identifies which animal group each influential parameter belongs to so that reduction efforts can be targeted at the specific sub-category that drives uncertainty rather than at the parameter in aggregate. The same labels appear in the rank-correlation table and the downloadable Word / Excel sensitivity sheets.

The tornado chart displays the top parameters ranked by |SRC|:

- **Green bars** (positive coefficient): increasing this parameter increases total emissions; reduction in uncertainty would directly reduce total emission uncertainty.
- **Red bars** (negative coefficient): increasing this parameter decreases total emissions.
- **Dark (slate) bars**: IPCC-defined coefficients that cannot be improved by the national inventory compiler (e.g. GWP values, Cfi, Cp). Uncertainty in these parameters is irreducible without IPCC guidance changes.
- **Bright (green/red) bars**: user-reducible parameters where better national data could meaningfully reduce inventory uncertainty.

# 8 Correlation Structure

## 8.1 Activity Data Correlations

Correlations between activity data parameters (e.g. body weight and mature weight, milk yield and milk fat content) are entered in the Tab~4 *Activity-Data Correlations* card. Four modes are available; each is described in detail below.

- **No correlations** (independence): the default. All parameters are drawn independently. This is the conservative IPCC Approach 2 starting point and is appropriate when the inventory compiler has no defensible information about parameter co-movement.
- **From template (auto, time-series)**: the matrix is computed automatically from the `Parameter_TimeSeries` sheet of the upload using a **Spearman rank correlation** after detrending (first differences by default; linear and raw alternatives available). Spearman is preferred over Pearson because (i) it is robust to skewed marginals (Ym, Bo, milk yield), (ii) it matches the rank-correlation target the sampler reproduces (per IPCC Vol.1 Ch.3 §3.2.3.2), and (iii) it is the preferred option when the user wants the *rank* dependence preserved rather than a linear product-moment relationship that can be distorted by outliers. Requires $\geq 5$ annual rows and $\geq 2$ non-constant numeric columns to produce a usable matrix; otherwise the option is greyed out (see "UI gating" below).
- **Structural-defaults preset**: a small set of literature-based pairs drawn from the IPCC equation structure and the published livestock literature (see Section "Structural-Defaults Preset" below). Independent of the upload's time-series, so this is the option that works for single-year inventories without historical data.
- **Manual entry**: the user uploads a square symmetric CSV. The diagonal must be 1; entries are interpreted as Spearman rank correlations on $[-1, +1]$; the matrix is projected onto the nearest positive-definite correlation matrix if needed, with a warning if any entry shifts by more than 0.01.

*UI gating.* Modes whose prerequisites are missing are greyed out automatically and cannot be selected: `From template'' is disabled when the upload has no usable \texttt{Parameter\_TimeSeries} sheet,`Manual entry'' is disabled until the user uploads a CSV matrix, and `Structural defaults'' is disabled until a template (built-in example or custom upload) is loaded. The Run button additionally blocks the simulation with an explicit error if a no-op mode somehow slips through. These gates prevent the silent fall-back to`no correlations'' that would otherwise occur when a mode is selected without the data it needs.

*Expected magnitude of effect on the headline.* Picking the structural-defaults preset on a typical single-year inventory changes the 95% uncertainty interval by about one percentage point or less (built-in examples at 3,000 iterations: Country X 25.3% without correlations and 25.2% with the preset; Country Y 36.5% and 37.5%). An earlier version of the preset, with DE $\leftrightarrow$ Ym at $-0.50$,

widened the interval by 3–10%; the October 2026 revision lowered that pair to −0.20 (see the Structural-Defaults Preset section below). The central estimate is essentially unchanged because correlations preserve marginal means. The "From template (auto)'' mode produces effects in the same general range when the time-series is rich enough to deliver multiple correlated pairs; a sparse or near-static time-series will produce a much smaller effect (the matrix collapses to near-identity after detrending).

## 8.2 Emission Factor Correlations

Coefficients are sampled independently of one another. This is the standard IPCC Approach 2 assumption and, since October 2026, the tool's unconditional behaviour.

Earlier versions offered a block-structured mode in which coefficients were grouped into three blocks (energy-equation, manure-$CH_4$, manure-N), each carrying its own within-block $\rho$ in $[0, 0.5]$, with cross-block correlation fixed at zero. The mode was withdrawn on two grounds. First, no inventory dataset supplies a within-literature correlation coefficient, so the control required an input the compiler could not evidence. Second, measurement showed most of it to be inert: on a 32-group inventory at 3,000 iterations, moving all three sliders from 0 to 0.5 moved the 95% margin of error from 9.09% to 11.00%, of which the energy block contributed −0.05 points and the manure-$CH_4$ block +0.05 points, both within sampling noise. The manure-N block accounted for the entire 1.9-point span, which follows from its parameter widths: EF3_PRP at ±221%, Frac_LEACH_PRP at ±150%, EF5 at ±89% and Frac_GASM_PRP at ±73%, against ≤ 30% for every coefficient in the other two blocks.

Correlation between coefficients remains expressible where it is evidenced. The unified correlation matrix spans both activity data and coefficients, so the structural-defaults preset (which includes the DE–$Y_m$ pair) and a user-supplied manual matrix both act on coefficients. What was removed is the block abstraction and its sliders, not the capability.

## 8.3 Structural-Defaults Preset (Literature-Based)

The tool provides a **structural-defaults preset** correlation matrix as a middle ground between full independence and user-specified correlations. It assigns a non-zero rank correlation to seven pairs of parameters that are linked through the IPCC equations or through well-documented biology; all other pairs are zero. *The values are literature-based: the direction of each link comes from the IPCC equations and from published studies, and the size of each value was set by the project team from the evidence listed below. The IPCC publishes no numerical correlation coefficients for any livestock parameter pair.* Hover any non-zero cell in the heatmap for the per-pair source. The preset is appropriate when no empirical correlation data are available but the inventory compiler wants to avoid the unrealistic assumption of full independence.

The current preset contains seven pairs:

| Pair | $\rho$ | Biological reason | References and status |
|---|---|---|---|
| BW ↔ MW | +0.50 | Live weight and mature weight enter the net energy for growth together (Eq 10.6, through $BW/(C \cdot MW)$), and animals of larger breeds are heavier at every age and have a larger mature weight. Set below the same-survey value (about 0.85) because MW is usually a breed-reference constant while BW comes from the census. | IPCC (2006) Eq 10.6; IPCC (2019) Table 10A.2 (r = 0.69 across 17 growing-cattle rows paired with the regional mature-cow weight). No published coefficient; project judgement. |
| BW ↔ WG | +0.40 | Both enter Eq 10.6 ($NE_g = 22.02 \cdot (BW/(C \cdot MW))^{0.75} \cdot WG^{1.097}$); well-fed animals are both heavier and gain faster. The sign reverses for young stock approaching mature size. | Mehrban et al. (2021): phenotypic correlation of daily gain with body weight 0.49–0.63 in 15,279 Hanwoo cattle; IPCC (2019) Table 10A.2 growing categories 0.2–0.5. Literature-based. |
| Milk ↔ Fat | −0.30 | Dilution effect: as milk yield rises, fat yield rises less than proportionally, so the fat percentage falls. | Boujenane (2002): −0.28 phenotypic in 20,140 Holstein-Friesian lactations; Nistor et al. (2011): −0.26 in 120 cows; IPCC (2019) Table 10A.1: −0.30 across nine regions. Literature-based. |
| Milk ↔ BW | +0.30 | Larger dairy breeds and more intensive systems have both heavier cows and higher milk yields; within one breed the link is weak. | IPCC (2019) Table 10A.1: 0.84 across regions; Berry and Evans (2022): the heaviest quarter of 16,980 Holstein-Friesian cows yields only 3–4% more; Berry et al. (2003): genetic correlation close to zero. Project judgement between the across-breed and within-breed cases; a single-breed herd may use 0.10–0.20. |
| Milk ↔ DE | +0.30 | More digestible feed gives more net energy per kg of feed and supports higher milk yield. | NRC (2001) Ch. 2; IPCC (2019) Table 10A.1: 0.65 across regions; no within-system coefficient is published. Project judgement, conservative. |

| Pair | $\rho$ | Biological reason | References and status |
|---|---|---|---|
| DE ↔ CP | +0.20 | Within one forage, digestibility and crude protein fall together as the plant matures. Across different feeds the link is weak, because protein and digestibility depend on different things (plant family, plant part, processing). | INRAE-CIRAD-AFZ (2018): $r = -0.07$ over 193 plant feed materials, and energy digestibility 57–64% across the dehydrated-alfalfa protein grades; Feedipedia: 0.1–0.2 over 82 forages with measured digestibility. Project judgement near the measured across-feed value; use 0.3–0.5 when the uncertainty is mainly about forage maturity. |
| DE ↔ Ym | −0.20 | IPCC Table 10.12 assigns a lower Ym to more digestible diets, mainly because high digestibility goes with high feeding level and grain-rich rations, which shift rumen fermentation and speed up passage; within forage-based diets, methane per unit of energy changes little with digestibility or rises slightly. | IPCC (2019) Table 10.12 (implies $r \approx 0$ under the Country X ranges, −0.32 read as a straight line); Ramin and Huhtanen (2013): positive link on forage diets (298 diets); Johnson and Johnson (1995). System-dependent: −0.20 for forage-based systems; −0.3 to −0.5 for high-grain dairy or feedlot systems via the manual matrix. |

*Revision history.* May 2026 audit: a previously included N ↔ BW = 0.30 pair was removed (N is sampled as a scalar broadcast, so the pair was inert), and Cfi ↔ Ca was lowered from 0.60 to 0.30. June 2026 livestock-science review: BW ↔ MW lowered from 0.85 to 0.50 (mixed-source provenance); Milk ↔ BW and Milk ↔ DE added as cross-group biological linkages; Milk ↔ pct_pregnant dropped (the sign is ambiguous between per-cow genetic and per-herd management interpretations) and Cfi ↔ Ca dropped (independent inputs in Eq 10.3 / 10.4, not jointly estimated). October 2026 evidence run: the label changed from "expert-elicited" to "literature-based"; DE ↔ CP lowered from +0.50 to +0.20, because the previously cited "cross-feedstuff correlations of 0.4–0.7" appear in none of the named sources and measured feed tables show no across-feed link; DE ↔ Ym lowered from −0.50 to −0.20, because the previously cited Eq 10.21 is the emission-factor equation rather than a Ym–DE relation, Table 10.12 implies a near-zero correlation under typical input ranges, Niu et al. (2018) does not use digestibility as a predictor, and measured forage-diet data give a positive link; citations that could not be verified (Brody 1945; Wilmink 1987; VandeHaar 1998; Stallings and Knowlton 2013; the "NRC 2001 joint regression") were removed from all seven sources. The resulting matrix is positive-definite without nearest-PD repair (smallest eigenvalue 0.26).

## 8.4 Cross-Block Correlations

The preset and the manual / time-series matrices support cross-block correlations (i.e. correlations between an activity-data parameter and a coefficient: for example BW ↔ Cfi). This is implemented via a single unified correlation matrix that spans the union of all parameter names; the sampler draws

from one multivariate distribution covering both blocks rather than two independent block-level distributions. Independence within and across blocks is the default.

## 8.5 Time-Series Correlation Matrix

When the user selects "From template'' with a `Parameter_TimeSeries` sheet loaded, the tool starts from the seven structural pairs of the previous section and lets the country's own series set the size of a pair, under three conditions: the series has at least ten years; the Spearman rank correlation of the year-to-year changes of the two parameters (first differences, which remove the shared long-run growth that is not the joint uncertainty Approach 2 is meant to capture) is different from zero at $p < 0.05$; and its sign agrees with the literature. A pair that fails a condition keeps its literature value. No other pair is created. The matrix is then projected onto the nearest positive-definite correlation matrix.

The restriction (30 October 2026) replaces an earlier behaviour that applied a correlation between every two columns of the sheet. Two reasons. With $n$ years there are $n - 1$ changes per parameter, and with five years any pair can show a rank correlation near $\pm 0.9$ by chance. And the co-movement of national annual values measures shared drivers (weather, policy, data revisions), which is not the correlation between estimation errors that IPCC Approach 2 asks for; the literature pairs describe the mechanism, which is closer to it. The Correlations tab shows the check pair by pair: literature value, value in the series, number of years, $p$-value, and which one the run uses. When no pair passes, the run uses the structural defaults and the tab says so.

## 8.6 Across-Group Coefficient Correlation

An inventory is normally disaggregated into groups (cattle type × production system × animal sub-category). A second decision then arises, exactly parallel to the year-to-year decision described in the next section: within a single Monte Carlo iteration, when two groups use the *same* value for a coefficient, should they receive the same draw or separate draws?

The question is not cosmetic. Uncertainty in a shared estimate is **systematic**. If the true $Y_m$ is 7.0 rather than the 6.5 applied to every group, every group is wrong in the same direction, by the same relative amount, simultaneously; nothing cancels in the sum. Drawing it independently per group asserts instead that the published figure is re-drawn afresh for each sub-category, and the resulting errors partially cancel. Because the cancellation goes as $1/\sqrt{N}$, the reported uncertainty then falls as the inventory is split more finely.

The magnitude is large. Holding the herd, the parameters and the uncertainty inputs fixed and varying only the number of groups:

| Groups | Mean (t $CO_2$eq) | 95% MoE |
|---|---|---|
| 1 | 1,188,109 | 25.1% |
| 4 | 1,189,861 | 12.5% |
| 16 | 1,188,003 | 6.3% |
| 64 | 1,188,164 | 3.1% |
| 251 | 1,188,033 | 1.5% |

The mean is stable throughout; only the uncertainty moves, and it moves purely with a reporting choice.

IPCC 2019 Refinement Vol.1 Ch.3 addresses this directly. It warns that subcategories `can be highly correlated, because either the AD are derived from the same source or the EFs have parameters in common''; it states that`in the case of full correlation among categories, aggregation of these categories is the recommended procedure'‘; and it recommends Approach 2 `where partial correlations are known to exist and are relevant''. It also gives the general rule, in the context of the time dimension:`emission factors (and other estimation parameters) uncertainties will tend to be correlated between years while activity data will tend to be uncorrelated between years.''

*Status in the app (30 October 2026).* Every run uses the shared draws described below, in single-year and in trend mode; there is no setting. The rule is stated on the Correlations tab and in the user guide. (Between 28 and 30 October the app had sampled every group independently.)

**The rule the tool applies when shared draws are used.** Groups carrying an identical estimate for a coefficient share a single draw; activity data stays independent. Formally, for each coefficient the groups are partitioned by their (mean, lower bound, upper bound, distribution) tuple; every group within a partition receives the same per-iteration draw, and separate partitions are drawn independently. Population $N$ is never shared.

Three properties follow:

- **Shared IPCC defaults are handled automatically.** Every group that fell back to the catalogue carries the identical value, forms one partition, and is therefore fully correlated, which is the only defensible treatment of a single published number.
- **Block structures are handled by the same mechanism.** An inventory with one value per agro-region forms one partition per region: correlated within a region, independent between regions, with no configuration.
- **It is self-correcting.** A compiler who genuinely estimated a coefficient separately per group has entered different numbers, so those groups land in different partitions and remain independent. An identical value is the evidence of a shared source.

Partitioning on the full tuple also guarantees that the marginal distributions within a partition are

identical, which is the precondition for sharing sampled values directly rather than sharing ranks.

**Shared draws and the correlation matrix (30 October 2026).** A shared draw is one number given to every group in its partition, so it cannot follow a different partner draw in each of those groups: one $Y_m$ handed to twelve groups cannot be low where a group's own $DE$ is high and high where it is low. The active correlation matrix is therefore applied between shared estimates only where they cover exactly the same set of groups (one $DE$ and one $Y_m$ for the same twelve groups move together as the matrix says, with the same restricted-pairing procedure and unchanged marginals), and between a group's own values among themselves, as before. A pair made of a shared estimate and a group-specific value is left uncorrelated. In trend mode the same rule is applied to the coefficient blocks drawn for the whole series (`fully correlated'') and to the normal innovations of the AR(1) chain` (partial'), so the matrix reaches the series as well as the single-year run.

**Verification.** Because IPCC states that full correlation should be equivalent to aggregation, the implementation has a hard acceptance property: splitting an inventory into more groups must not change the reported uncertainty. With sharing enabled the table above becomes approximately flat (25.1% on one group, 22.7% on 251). The residual decline is the genuinely independent activity-data component averaging out, and is quantitatively consistent with the 10% uncertainty placed on $N$ in that test: $\sqrt{25.1^2 - 22.7^2} \approx 10.7\%$.

**A real inventory.** The same comparison on a real national Tier 2 inventory (32 groups, 800 parameter rows, 4,000 draws, no activity-data correlations). The mean is unchanged (9.25 Mt $CO_2$eq); the activity-data-only margin is identical (12.9%) because population is never shared; the whole difference sits in the emission-factor side, where Ym (one value in 12 groups), Cfi (one value in 22) and the pasture $N_2O$ factors (one value in all 32) are drawn once instead of 32 times. Cutting every group into two and four identical parts, same animals and $N$ divided, reproduces the Country X pattern.

| | **Separate draws** | **Shared draws** |
|---|---|---|
| Total $CO_2$eq, 95% MoE (32 groups) | 23.7% | 32.5% |
| Enteric $CH_4$ | 24.3% | 33.4% |
| Pasture $N_2O$, direct / indirect | 39.7% / 58.2% | 103% / 148% |
| AD-only / EF-only | 12.9% / 19.2% | 12.9% / 29.2% |
| Same herd in 64 groups | 16.8% | 30.9% |
| Same herd in 128 groups | 11.5% | 30.3% |

In the sensitivity ranking the shared estimates appear once each, labelled with the value and the groups they cover ("DE = 52.8 (extensive beef, 6 groups)''); with separate draws the same uncertainty is spread over one bar per group and the ranking is dominated by the largest groups.

**Scope.** Only full sharing is implemented. Partial sharing ($0 < \rho < 1$), appropriate where a shared

systematic component coexists with independent per-group sampling error, is not offered: it requires an $n_{\text{iter}} \times n_{\text{groups}}$ matrix per parameter held simultaneously, which is not feasible within the hosting memory budget at the group counts where the issue matters most. IPCC positions partial correlation as a refinement to apply "where known to exist and relevant''.

## 8.7 Animal Numbers Between Groups and Between Years

Population $N$ is never shared between groups by the rule of the previous section (each group's own count is its own estimate), yet in most inventories the groups' numbers are shares of one national count, and in a time series the same count, interpolated, serves many years. Drawing each group's $N$ on its own, afresh every year, lets a census error cancel between groups and between years. IPCC 2019 Refinement Vol.1 Ch.3 names activity data "derived from the same source'' as a case of correlated sub-categories, and the 2006 Guidelines present the independence of activity data between years (Approach 1, Eq. 3.2) as a simplifying assumption that Approach 2 can relax where the same source serves both years.

Since 30 October 2026 the Correlations tab asks two questions about $N$ and the sampler honours them. Normal scores $z$ (one column per group) carry the structure: between groups, $z_g = \sqrt{\rho}\, c + \sqrt{1-\rho}\, \varepsilon_g$ with a common $c$ and $\rho = 1$ (one count split into groups, the preselected answer), 0.5 (partly) or 0 (separate counts); between years, in trend mode, the scores are kept for every year (one census, interpolated), follow an AR(1) with $\rho = 0.7$ (a census every few years) or are renewed each year (annual survey, the preselected answer). Each group's own marginal distribution is then placed on the ranks of its column, so means and ranges are unchanged; only the way the draws are tied together changes. The EF-only run of the decomposition drops these draws, as it holds $N$ at its mean. In the sensitivity ranking a count split into groups appears as one bar ("N (one count, 32 groups)''), the sum of the groups' draws, because their ranks are identical and a regression cannot separate them.

**A real national inventory (32 groups, no activity-data matrix, shared coefficient draws).** Single year, 4,000 draws: total 95% MoE 32.5% with separate counts, 40.1% partly, 47.1% with one count split; AD-only 12.9%, 27.1%, 36.5%; EF-only 29.2% throughout; the mean unchanged. Trend 1990 to 2024, 2,000 draws per year, coefficients fully correlated between years:

| Between groups | Between years | 1990 MoE | 2024 MoE | 95% range of the trend (pp) |
|---|---|---|---|---|
| separate counts | annual survey | 34.8% | 35.7% | −61 to +78 |
| separate counts | census every few years | 35.2% | 35.7% | −62 to +82 |
| separate counts | one census, interpolated | 35.2% | 35.4% | −3 to +2 |
| partly | annual survey | 48.1% | 48.8% | −110 to +201 |
| partly | one census, interpolated | 48.1% | 47.7% | −3 to +4 |
| one count split | annual survey | 58.3% | 58.6% | −140 to +315 |
| one count split | census every few years | 58.3% | 59.5% | −138 to +314 |
| one count split | one census, interpolated | 58.3% | 57.9% | −3 to +4 |

The between-groups choice moves each year's margin; the between-years choice moves the trend's: a census error kept for every year cancels in the comparison of two years, a renewed one does not.

### 8.8 Year-to-Year Coefficient Correlation (Trend Mode)

When the tool is run in **Trend mode** (multi-year inventory), an additional decision arises: within a single Monte Carlo iteration, how should the IPCC coefficients (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5, ...) move from one year to the next? IPCC Volume 1 Chapter 3 Section 3.2.3.2 (Trend Analysis) and 2019 Refinement §3.2.2.4 discuss this explicitly: emission factor uncertainties are typically estimated once for an inventory series and reused across years (correlated), while activity data is normally re-estimated each year and treated as independent across years.

The tool exposes three options on Tab 3 (Correlations) when Trend mode is selected. Only the animal numbers ($N$) are activity data; their link between years is a separate choice on the same tab. Every other input (BW, WG, Milk, DE and the IPCC coefficients) follows this setting. Each year's draws come from that year's own values and ranges in the time series; the setting only fixes how the errors of two years are linked:

| Option | What happens inside one Monte Carlo iteration |
|---|---|
| **Fully correlated (default)** | A coefficient keeps the *same rank* in every year of one iteration: a draw at the 80th percentile of its range in year 0 is at the 80th percentile in every other year. A value that does not change over the years gives the identical draw (if Ym is drawn at 6.2% for year 0, it is 6.2% for every year); a value the time series changes (body weight, milk yield, digestibility) is drawn around the new value at the same relative position. The coefficient uncertainty then cancels when two years are compared, and the trend reflects the changes in the values and in the activity data. Matches IPCC 2019 §3.2.2.4 default behaviour and the Ogle et al. (2003) "identical random seed" construction cited in V1 Ch3 Box 3.2. |
| **Partial (AR(1), $\rho = 0.7$)** | Coefficients drift slowly between years: the ranks follow a Gaussian AR(1) with $\rho = 0.7$, a rank correlation of about 0.68 between neighbouring years, falling with the distance between years. Each year's draws come from that year's own marginal and are reordered by the AR(1) ranks. Neighbouring years share most of the same observational basis but the coefficient can move over a decade. Appropriate when the EF is re-estimated periodically (e.g. every 3–5 years). |
| **Independent** | Each year gets a fresh, independent coefficient draw. Maximises the EF contribution to trend uncertainty. Realistic only when every emission factor is genuinely re-measured with new field data every year, which is uncommon for national inventories. |

The choice is implemented in `make_year_block_generator()` (`R/systems_builder.R`). Each coefficient draw keeps one latent normal vector of length $n_{\text{iter}}$. Each year, the coefficient's marginal is drawn from that year's own value and range, sorted, and reordered by the rank of the latent. Under `full'' the latent is the same every year, so the rank correlation between any two years is 1. Under`partial'' it follows a Gaussian AR(1), $z_t = \rho z_{t-1} + \sqrt{1-\rho^2}\, e_t$, whose lag-$k$ Spearman correlation is $(6/\pi)\arcsin(\rho^k/2)$, about 0.68 at lag 1 for $\rho = 0.7$. The `independent'' branch re-samples coefficients inside every per-year call. Until 4 November 2026 the`full'' and "partial'' branches drew the coefficients once from the inventory-year values, so a time series of body weight, milk yield or digestibility did not reach the yearly means under those two settings; on a national test inventory the correction lowered the 1990 total by 5% and raised the

1990 to 2024 change from +152% to +164%.

**Draws per year, and the two reported years.** The series runs every year at the trend iteration count (2,000 by default), which is enough for the chart and the slope: on a national inventory of 32 sub-categories the 95% margin of error of a year is known to about ±1 percentage point at 2,000 draws against ±0.6 at 10,000, and the standard error of the mean is 0.3% against 0.13%. The headline figures and IPCC Table 3.3 depend on the base year and year $t$ only, so after the series those two years are simulated again, as a pair, at the reporting draw count (10,000 by default, the single-year setting): the same coefficient ranks in both years under `fully correlated'', the lag-$k$ AR(1) link under`partial'', independent under "independent", each year with the seed the series gives it. The activity-data and emission-factor columns of Table 3.3 (further simulations with the other side held at its mean) are run at the same count on the same systems, for both years since 4 November 2026: the year-$t$ columns give the uncertainty of year-$t$ emissions from each side, and two trend columns give the uncertainty each side introduces into the trend, the 95% half-width of the per-draw change of the activity-data-only and emission-factor-only pairs, in percentage points. They are the Monte Carlo counterpart of columns K and L of the Approach 1 table (IPCC 2006 Vol. 1 Ch. 3 Table 3.2); being separate runs, they need not combine in quadrature to the combined trend uncertainty. Run at the series' own count, the paired run reproduces the series' draws for the two years exactly, so it is the same computation at higher precision. Choosing two other years on the report tab runs them the same way.

**Direction of effect.** For an emission factor whose uncertainty is largely systematic, full correlation produces the *smallest* trend uncertainty (because the EF cancels) and independent sampling the *largest*. IPCC V1 Ch3 p. 25 explicitly notes that `positive correlations in uncertainties when comparing two years as part of trend analysis will decrease uncertainty in the trend''. Choosing`independent'' when the EFs are in fact reused will over-state trend uncertainty; choosing `full'' when the EFs are in fact re-measured each year will under-state it. The IPCC default (`full'') is the conservative choice for the typical case where the inventory compiler is using IPCC defaults or a single national estimation programme.

**Inventories whose systems differ.** The setting applies to every group. When some production systems keep the same factors for the whole series and others revise them, or only some factors are revised, the setting that fits the systems with the largest share of emissions is the natural choice, and a second run with the other setting bounds the result: the larger of the two trend uncertainties is the conservative figure to report, or both are reported with the reason in the methods section of the national inventory report. "Partial'' lies between the two and is a reasonable single choice for a mix.

**Sensitivity of the change.** The ranking of the inputs that drive the uncertainty of the change regresses the per-draw change on the inputs of both years. Inputs that move together are one input for this regression: columns of one parameter whose rank correlation exceeds 0.999 (animal numbers drawn as one count split into groups, or a count or coefficient linked between the two years) are merged, the animal numbers into their sum. Inputs that cannot move a group's emissions (the factors of a manure system with a zero allocation, constants) are left out, and when more inputs

remain than one per ten draws, those with the strongest rank correlation with the change are kept (at least 50). Without these steps the regression held more inputs than draws and its standardised coefficients were meaningless.

**Results by cattle type.** IPCC Table 3.3 is also given separately for dairy and non-dairy cattle, in single-year and trend runs. Each type's figures come from the per-draw results of its own groups, summed within the type before the percentiles are taken, so the shared draws between groups are kept; no extra simulation is run.

# 9 Validation and Quality Assurance

The tool includes a QA/QC module (`R/utils_qaqc.R`) that runs a series of checks against the loaded parameter data before simulation. Each check returns one of five statuses:

| Status | Trigger | What the user sees |
|---|---|---|
| **Pass** | All structural checks pass. For `BW`, the mean is also within ±50% of the continental IPCC default (Annex Tables 10A.1 / 10A.2 / 10A.3). | Green tick; no action needed. |
| **Warn** | For `BW`: value deviates 50–200% from the continental IPCC default. For any parameter: implausible bounds ($BW > 1500$ kg, $Y_m > 12\%$); a fractional parameter is using an unbounded distribution; or another non-fatal structural condition fired. | Amber row; the message either suggests a fix or asks the user to document the country-specific deviation. |
| **Fail** | For `BW`: value deviates more than 200% from the continental IPCC default. For any parameter: $DE_{\%} \leq 0$ or $\geq 100$; or $lower > upper$; or an MMS fraction-sum is more than 1% off 100%; or a Beta is configured with mean outside $(0, 1)$. | Red row; simulation will still run but the result will be flagged in the report. |
| **Missing** | A core parameter was not supplied in the upload; auto-filled with the IPCC default. Applies to every parameter, not just `BW`. | Grey row; the message names the IPCC table or equation used and, for context-dependent parameters (EF3_PRP, EF4, Frac_LeachPRP, Bo, MCF, Ym, Cfi, Ca), appends an explicit hint of which alternative IPCC table row should be reviewed for the inventory's climate / production system / animal class. |
| **Info** | Reserved for future advisory messages. | Blue row. |

The Pass / Warn / Fail thresholds for benchmark deviation apply **only to body weight (`BW`)**, the single parameter for which the tool holds a defensible continental IPCC default lookup: Vol.4 Ch.10 Annex Table 10A.1 (dairy cows), Annex Table 10A.2 (other cattle), or Annex Table 10A.3 (buffalo). Earlier versions of the tool also flagged deviations on Milk, MW, DE, Ym, and Bo against heuristic mid-points derived from the same Annex 10A illustrative tables; these were removed (Andreas Wilkes review, May 2026) because those mid-points were not directly citable to a published IPCC default and a reviewer auditing the QA tab could not locate them in the guidelines. The `ipcc_default` column of `PARAM_CATALOGUE` is still used to auto-fill missing values and is sourced from Vol.4 Ch.10 (Tables 10.4, 10.5, 10.7, 10.11, 10.12, 10.16(a), 10.17, 10.21, 10.22, 10.23) and Vol.4 Ch.11 (Tables 11.1, 11.3), it is just no longer used as a deviation benchmark for parameters other than BW. Reinstating per-parameter benchmarks properly would require a multi-dimensional table indexed by continent, IPCC version (2006 vs 2019R), production system, and animal sub-category; that is deferred to a future release. Auto-filled parameters are listed in a banner on Tab~1 immediately after data upload; they should be replaced with country-specific values wherever possible, because both the central estimate and the propagated uncertainty depend on them.

# 10 Verification of Calculation Accuracy

The previous section describes the checks the tool applies to the user's *inputs*. This section describes how the tool's *calculation engine*: the implementation of the IPCC equation chain and the Monte Carlo sampler, has itself been verified to produce correct numbers. Input validation and calculation verification are complementary: the former guards against bad data going in, the latter guarantees the equations are implemented correctly.

## 10.1 Equation-chain verification against manual calculation

The core of the verification is a fixed, fully worked test case. For a single representative animal with a complete set of fixed parameters, every intermediate quantity in the IPCC Tier 2 chain was computed independently by hand and in a spreadsheet, directly from the published equations: the net-energy terms ($NE_m$, $NE_a$, $NE_g$, $NE_l$, $NE_w$, $NE_p$), the energy ratios ($REM$, $REG$), gross energy intake ($GE$), the enteric emission factor ($EF_1$), volatile-solid excretion ($VS$), nitrogen excretion ($N_{\text{ex}}$), and all four manure- and pasture-$N_2O$ pathways, through to the inventory totals (total $CH_4$, total $N_2O$, and total $CO_2$eq under each GWP basis).

The tool's computed value for each of these quantities must match the independently calculated reference to a relative tolerance of $1 \times 10^{-4}$. These comparisons are encoded as automated assertions that run on every build, so any future change to the calculation code that altered a result: even slightly, would be caught immediately. This is the primary guarantee that the equations in Section "IPCC Tier 2 Emission Equations" are implemented exactly as written.

## 10.2 Cross-comparison with an independent Monte Carlo package

Beyond the single worked case, the tool has been exercised on real Tier 2 national cattle inventories and its results compared against two independent references: (i) manual re-calculation of the per-head emission factors and inventory totals, and (ii) an independent commercial Monte Carlo package configured with the same parameters, distributions and iteration count.

On these real inventories the tool reproduced the deterministic emission totals from the manual calculation, and produced a 95% uncertainty interval consistent with the independent Monte Carlo package to within the sampling variability expected between two stochastic runs. %% OPTIONAL: if you wish to quote a specific figure, insert it here, e.g. %% "(agreement within X% on the 95% margin of error for the manure-N2O pathway)". A regression guard retained in the automated test suite keeps the manure-$N_2O$ pathway within the range established by this cross-comparison, so a future code change cannot silently introduce a large (e.g. order-of-magnitude) error in that pathway.

## 10.3 Automated regression test suite

The two verification activities above are part of a broader automated test suite (91 checks at the time of writing) that is run before each release and deployment. The checks are organised into seven groups:

- **A: IPCC equation chain:** every intermediate and total against the hand-calculated reference case (Section "Equation-chain verification" above).
- **B: Monte Carlo sampler:** empirical means of the sampled marginals match their analytical means; realised Spearman rank correlations match the requested targets; the AR(1) year-to-year correlation structure behaves as specified.
- **C: Application routes:** emission-source filtering (all sources, enteric only, manure $CH_4$ only, $N_2O$ only), the three GWP bases (AR4 / AR5 / AR6), the correlation modes, and the AD / EF / combined uncertainty decomposition.
- **D: Trend (multi-year) pipeline:** the full / partial / independent year-to-year coefficient-correlation modes produce valid multi-year results.
- **E: Multi-sub-category aggregation:** per-sub-category emissions sum correctly to national totals and scale as expected with population.
- **F: Edge cases and validators:** missing-value handling, manure-management fraction bounds, lower/upper-bound consistency, zero-population handling, and the multi-system $N_2O$ calculations.
- **G: Outputs:** the Excel and Word/CSV downloads and the correlation-matrix round-trip produce well-formed, internally consistent files.

The full multi-sub-category inventory chain must pass all checks before the tool is published.

### 10.4 Verification of IPCC default values and uncertainty ranges

Every IPCC default value and uncertainty range pre-loaded in the tool: the energy-equation coefficients, the methane conversion factor $Y_m$, the maximum methane producing capacity $B_o$, the manure-system MCF values, and the $N_2O$ emission factors and volatilisation / leaching fractions ($EF_3$, $EF_4$, $EF_5$, FracGASM, FracLEACH), has been checked line-by-line against the IPCC 2019 Refinement source text (Volume 4, Chapter 10 and Chapter 11). This review confirmed the central values and corrected two pre-loaded uncertainty ranges ($B_o$ and the pasture direct-$N_2O$ factor $EF_{3,\mathrm{PRP}}$) to match the climate-specific ranges in Tables 10.16a and 11.1.

### 10.5 Default uncertainty ranges

The tool applies a default uncertainty range only where the literature gives one that applies to any country. These are the ranges IPCC publishes with its default values, and the expert-judgement ranges of Monni et al. (2007) for four IPCC equation coefficients that no country can measure; the UK agriculture inventory uses the same four ranges (Milne et al., 2014). Table 5 lists them.

Every other input describes the country's own data: animal numbers, weights, weight gain, milk yield and fat, pregnancy, feed digestibility and crude protein, MCF, and the remaining manure inputs. Its range reflects how those data were collected, so it must come from the country. IPCC 2019, Vol. 1, Table 3.6 gives ranges for some of these inputs, but for one national example (Italy);

Monni et al. (2007) give them for Finland's animal register. Neither applies to another country's data. When the country gives no range for such an input, the tool uses a placeholder so that the simulation can run: the values the tool used before October 2026 for the parameters, and for the manure inputs MCF ±25%, $EF_3$ a factor of 2, and the volatilisation and leaching fractions ±50%. A placeholder has no published source. It is reported as a warning on the QA/QC tab, listed in the Word report, and marked in the Excel and CSV exports, so that it is never mistaken for a default or for the country's own range.

The asymmetric IPCC ranges belong to the IPCC default value. They are applied only when the input carries that value; a country that uses its own value supplies its own range. The percentage ranges are applied around the value given, and the QA/QC tab notes each one so that a country with its own value can replace it.

**Table 5:** Inputs with a default uncertainty range.

| Input | Default range | Source |
|---|---|---|
| $Y_m$ | ±20% | IPCC 2019, Vol. 4, Table 10.12, footnote 3 |
| $B_o$ | ±15% | IPCC 2019, Vol. 4, Table 10.16A |
| $EF_{3,\mathrm{PRP}}$ | 0.0005 to 0.027 | IPCC 2019, Vol. 4, Table 11.1 (lower bound 0 in IPCC; a small positive floor is used) |
| $EF_4$ | 0.011 to 0.017 | IPCC 2019, Vol. 4, Table 11.3 |
| $EF_5$ | 0.0005 to 0.020 | IPCC 2019, Vol. 4, Table 11.3 |
| $\mathrm{Frac}_{\mathrm{GASM}}$ (pasture) | 0.005 to 0.31 | IPCC 2019, Vol. 4, Table 11.3 |
| $\mathrm{Frac}_{\mathrm{LEACH}}$ (pasture) | 0.01 to 0.73 | IPCC 2019, Vol. 4, Table 11.3 |
| $EF_3$ by manure system | a factor of 2 | IPCC 2019, Vol. 4, Table 10.21, footnote 24 |
| $\mathrm{Frac}_{\mathrm{GasMS}}$, $\mathrm{Frac}_{\mathrm{LeachMS}}$ by system | the Table 10.22 range, where IPCC gives one | IPCC 2019, Vol. 4, Table 10.22 |
| $C_{fi}$, $C_a$, $C$ | ±30% | Monni et al. (2007), Table I |
| $C_p$ | ±10% | Monni et al. (2007), Table I |

## 10.6 Reproducibility

Each simulation is driven by an explicit random seed, so any run can be reproduced exactly by re-supplying the same inputs and seed. The convergence guidance in Section "Recommended Iteration Count and Convergence": re-running with a second seed and checking that the 95% margin of error is stable, lets a user confirm that a reported result is converged rather than an artefact of a particular random draw.

# 11 Reporting

The results calculated by the tool are designed to align with IPCC uncertainty reporting guidelines (IPCC 2006 Vol 1 Ch 3). The outputs reported by the tool can be used in a hybrid Approach 1 / Approach 2 uncertainty analysis for the whole inventory (IPCC 2006 §3.2.3.3) by entering into Table 3.3 the results reported by the tool for cattle emission sources alongside Approach 1 results for other emission categories. The outputs produced by the tool can also be used as inputs to a full Approach 2 analysis of uncertainty for the whole inventory.

## 11.1 IPCC Table 3.3 Inputs

In the case of a hybrid Approach 1 / Approach 2, IPCC 2006 Table 3.3 (see also the IPCC 2019 MS Excel tool for Approach 1, and the IPCC 2019 Vol 1 Ch 3 addendum) requires the following inputs to characterise uncertainty of each emission source, each expressed as $\pm x\%$ of the inventory emission estimate:

- Activity-data (population) uncertainty (Column E);
- Emission-factor uncertainty (Column F);
- Combined uncertainty (Column G).

The tool reports all three figures for each cattle emission source via the IPCC Report tab and the Excel / Word downloads. The activity-data vs emission-factor split follows the convention adopted in this tool: AD = animal population (N) only; coefficient (EF) = the IPCC equation parameters that combine into the per-head emission factor (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5, Frac_GasMS, Frac_LeachMS, Frac_PRP, EF3_PRP, Frac_GasPRP, Frac_LeachPRP).

## 11.2 UNFCCC Common Reporting Tables Category Map

The tool can calculate AD, EF and combined uncertainties for one or more cattle categories. While country-specific classifications are allowed for, the UNFCCC Common Reporting Tables (CRTs) indicate reporting of emissions from cattle at the following levels of disaggregation:

- **3.A Enteric fermentation ($CH_4$)**
  - 3.A.1.a Dairy cattle
  - 3.A.1.b Non-dairy cattle
- **3.B Manure management** ($CH_4$; direct and indirect $N_2O$ combined)
  - 3.B.1 Cattle (3.B.1.a Dairy, 3.B.1.b Non-dairy)
- **3.D Agricultural soils** (direct and indirect $N_2O$)
  - 3.D.1.c Urine and dung deposited by grazing animals (direct $N_2O$)
  - 3.D.2.a Atmospheric deposition (multiple sources, including dung and urine deposited on pasture)
  - 3.D.2.b Nitrogen leaching and run-off (multiple sources, including dung and urine deposited on pasture)

Country-specific options in the CRTs may include reporting of emissions at the level of production systems, regions or cattle sub-categories.

### 11.3 Disaggregation Levels

To facilitate uncertainty analysis at these different levels of disaggregation, the default outputs in the tool are produced at three disaggregation levels concurrently:

- **Level 1**: disaggregated by cattle type (dairy, non-dairy), emission source (3.A, 3.B, 3.D) and gas ($CH_4$, $N_2O$). Available in the IPCC Report tab and the headline summary of the Word / Excel run-summary.
- **Level 2**: Level 1 plus a user-defined second level (production system, region) using the template's `cattle_type` and `region` fields. Available in the per-cattle-type results section of the Word run-summary.
- **Level 3**: Level 2 plus cattle sub-categories (dairy_cow, heifer, bull, etc.). Available in the per-(cattle type × emission source) breakdown of the Word run-summary and the per-sub-category CSV download from the IPCC Report tab.

A user-selectable toggle to choose the reporting disaggregation level for the whole tool is not implemented in this release; the three levels are produced concurrently in their respective output channels. A single-toggle option may be added in a future version after feedback from UNFCCC partners on the right default level.

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